Research Seminar

Group N

11. FLOW TOXICITY AND LIQUIDITY IN A HIGH-FREQUENCY WORLD

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Why toxic order flow matters

- **Market makers** (today mostly high-frequency firms) earn a tiny spread on many trades and avoid directional bets.
- They lose when the other side **knows more**. Order flow is **toxic** when it adversely selects them, often before they notice.
- When toxicity rises they widen quotes or **leave**. With no one absorbing orders, prices gap. The paper links the **6 May 2010 flash crash** to this.
- **Question:** can toxicity be measured in real time, from trades alone, early enough to act?



Method: from PIN to VPIN, measured in volume time

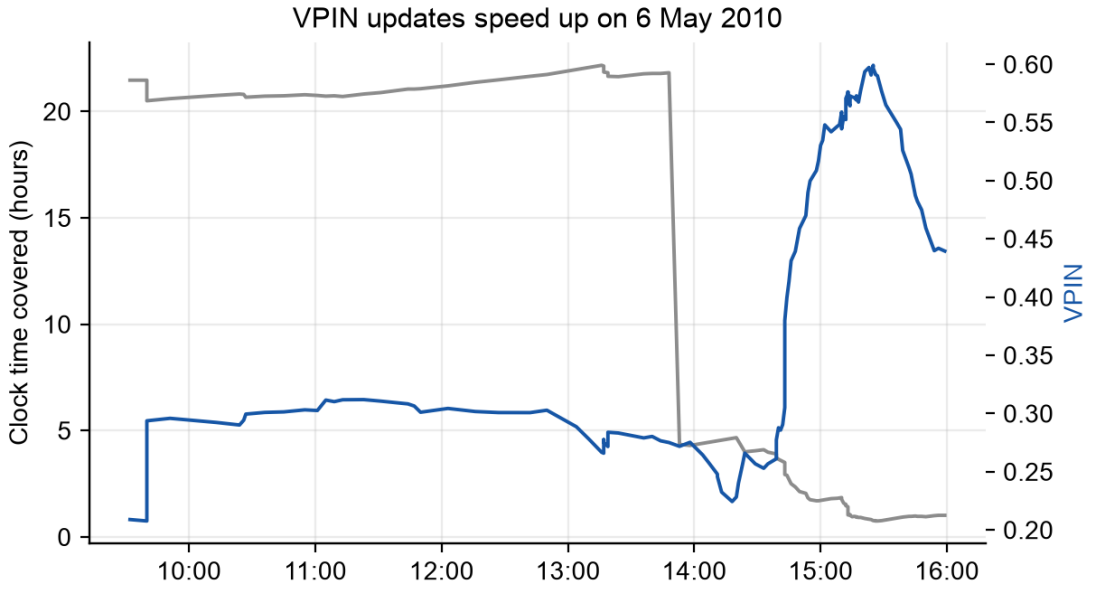
• **PIN** (Easley et al. 1996): the share of orders that come from informed traders

$$\mathrm{PIN} = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}$$

• α: chance of news, μ: informed arrivals, ε: uninformed arrivals per side. Needs maximum likelihood: slow and unstable at high frequency.

• **VPIN:** |buys − sells| estimates αμ, total volume estimates αμ + 2ε. No parameters to fit.

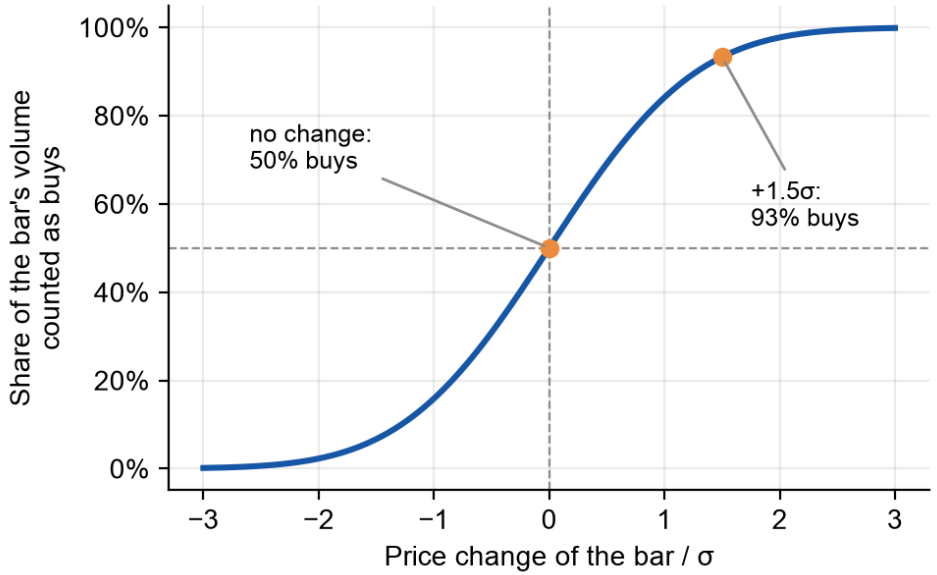
• **Volume clock:** cut trading into buckets of equal volume (1/50 of a day). Busy, news-heavy periods get more updates.



Method: bulk volume classification

$$V_\tau^B = \sum_{i \in \tau} V_i Z\left(\frac{P_i - P_{i-1}}{\sigma_{\Delta P}}\right), \quad V_\tau^S = V - V_\tau^B$$

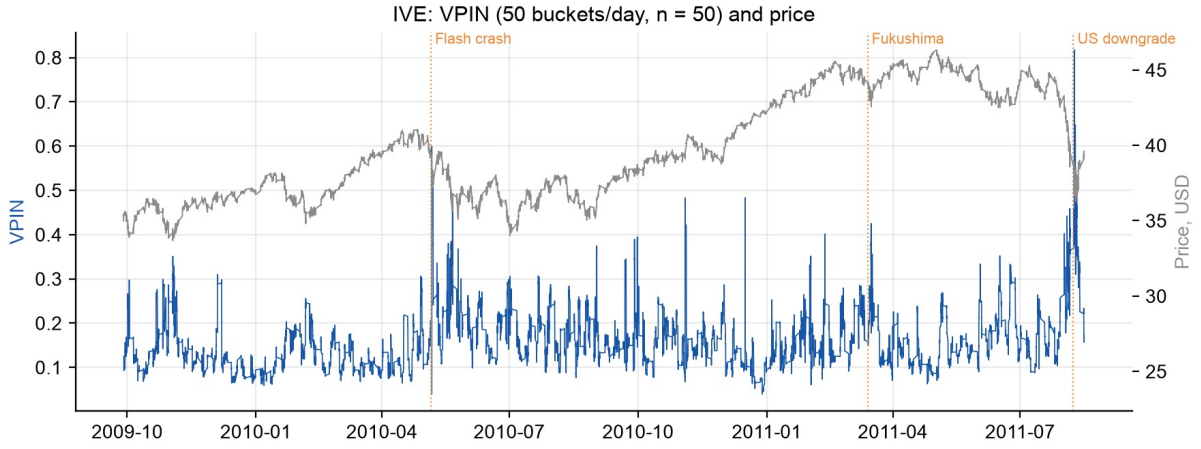
- Group trades into **one-minute bars**; split each bar's volume by its standardized price change.
- No price change → 50/50. A rise of 1.5σ → 93% buys. Z is the standard normal CDF; σ is the standard deviation of bar price changes.
- Robust to order splitting and cancellations, cheap to compute.
- **But:** the imbalance it produces grows with |price change|, by construction.



Method: the VPIN metric and how it is read

$$\mathrm{VPIN}_\tau = \frac{\sum_{j=\tau-n+1}^{\tau} \left| V_j^S - V_j^B \right|}{nV}$$

- Bucket size **V = average daily volume / 50**; window **n = 50** buckets ≈ one day.
- Updated after **every bucket**: on busy days many times per hour.
- Levels depend on classification and settings, so the paper reads **CDF(VPIN)** over the whole sample. Alarm: CDF > 0.9.
- Claim: high VPIN warns of **toxicity-induced volatility**.

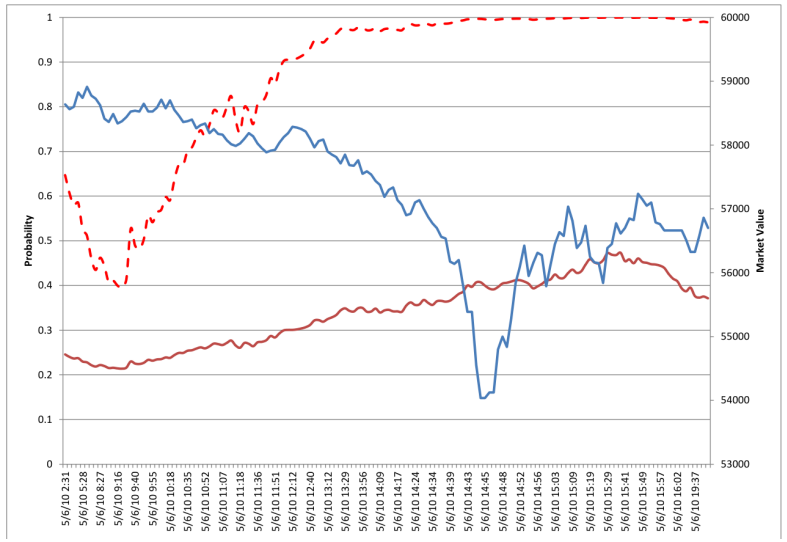


Data and our replication setup

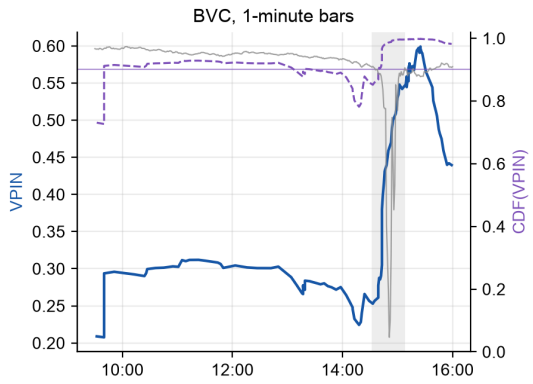
	Paper	Ours
Market	E-mini S&P 500 and WTI crude futures	IVE (S&P 500 Value ETF) ticks with quotes; BTCUSDT for extensions
Period	Jan 2008 – Aug 2011 (incl. 2008 crisis)	2009-09-28 – 2011-08-15; out of sample to Sep 2026
Trading hours	almost 24 hours	09:30–16:00 New York
VPIN observations (50, 50)	44,665	23,701
Classification	bulk, one-minute bars	same, plus 10-second bars and Lee-Ready
Settings	V = ADV/50, n = 50, whole-sample σ and CDF	identical

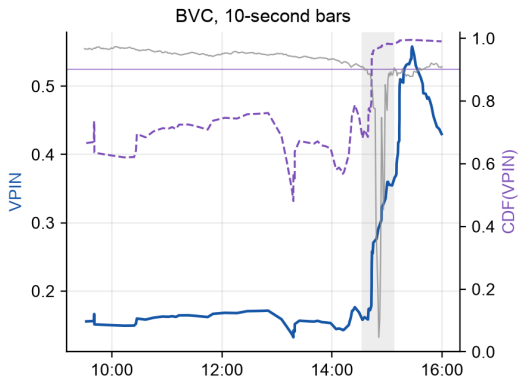
• Main deviations: ETF instead of futures; regular hours only; our sample starts in Sep 2009 (no 2008 crisis); 42 IVE trades at prices like $0.08 that the exchanges **cancelled** on 6 May 2010 are removed. 29% of IVE one-minute bars have no price change, and bulk classification splits them 50/50.

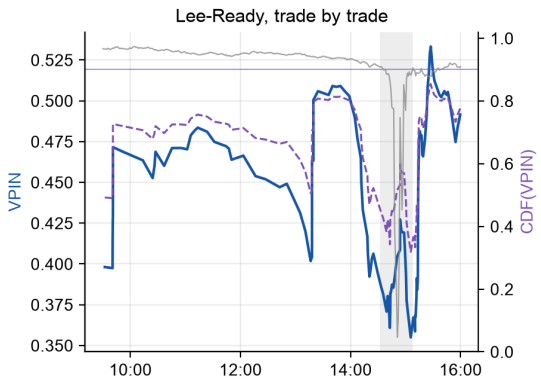
Result 1: the flash crash, paper vs ours



Paper, Fig. 4(a): E-mini, one-minute bars







- **One-minute bars:** CDF > 0.9 from 09:40, **4.9 h** before the fall — as in the paper.
- **10-second bars:** first above 0.9 at 14:43, during the fall. **Lee-Ready:** never (max CDF 0.85).
- The warning depends on the classification choice.

Result 2: the volume clock and the stability of VPIN

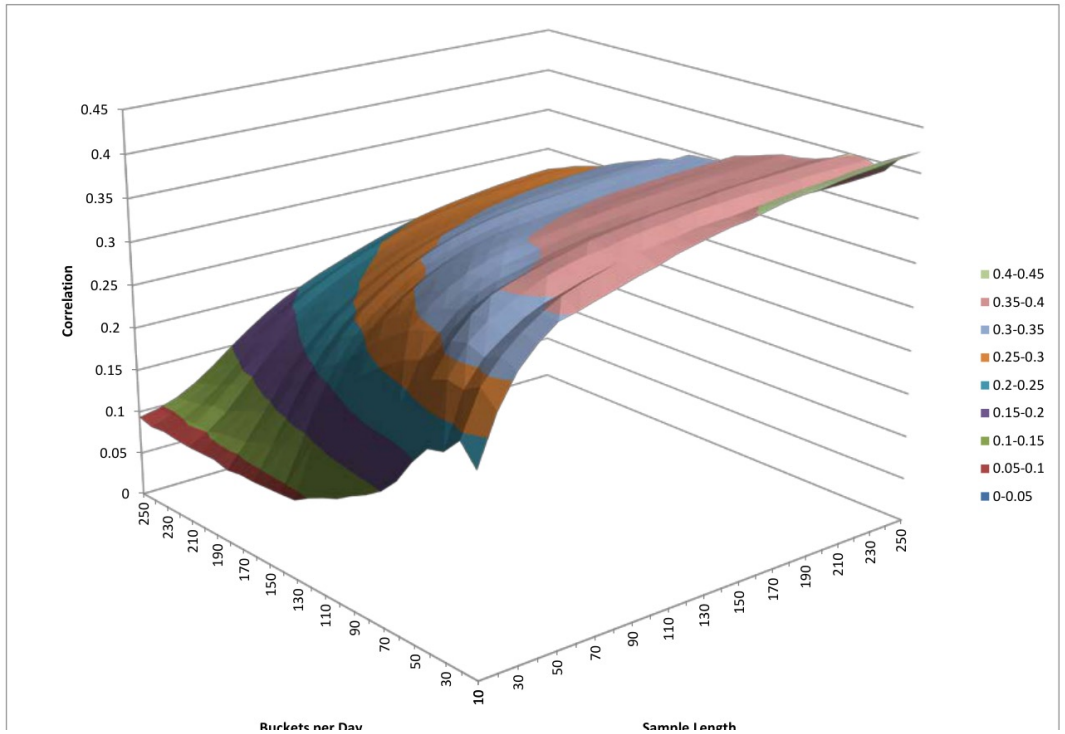
50 draws a day	Paper: clock	Paper: volume	Ours: clock	Ours: volume
Kurt	34.25	**20.52**	13.12	**9.05**
J-B*	40.69	**12.82**	4.27	**1.53**
L-B*	40.73	**24.84**	23.13	43.03
White*	0.10	**0.04**	0.28	**0.16**
Max	20.83	**12.26**	13.08	**9.86**

- Volume time improves 3 of 4 test statistics at 50 draws a day and 4 of 4 at 100 (green = lower than clock time).
- Computed without 6 May 2010 and without overnight changes; the raw table is dominated by the flash crash (DEVIATIONS.md, row 8).

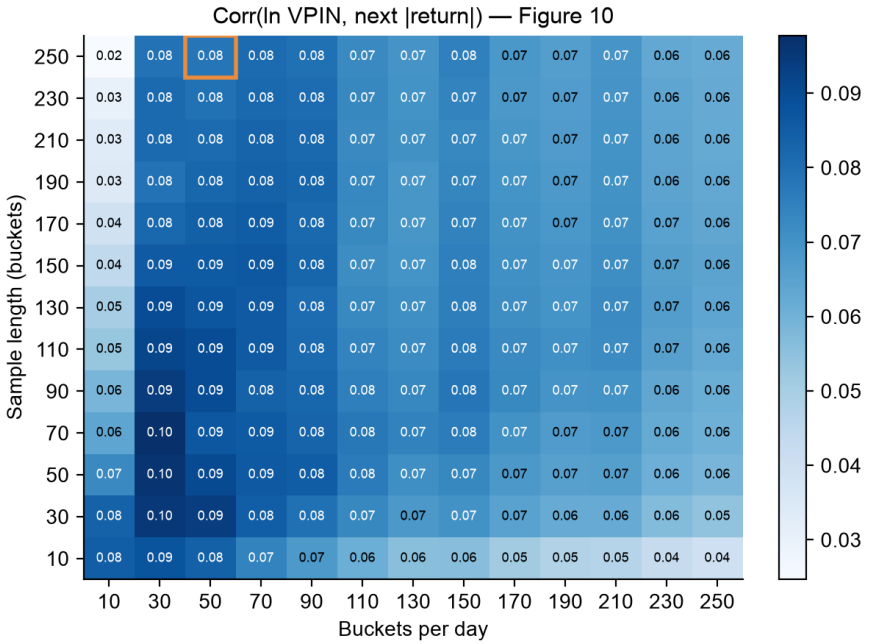
Stability (Fig. 5)

- Start the bucket sequence 0, 1, …, 999 bars later and compare the VPIN paths.
- Cross-sectional std / mean VPIN: paper **6.5%**, ours **3.7%**.
- VPIN does not depend on where the volume clock starts. **Reproduced.**

Result 3: does VPIN predict the next bucket's volatility?



Paper, Fig. 10: E-mini



At (50, 250)

paper **0.40**

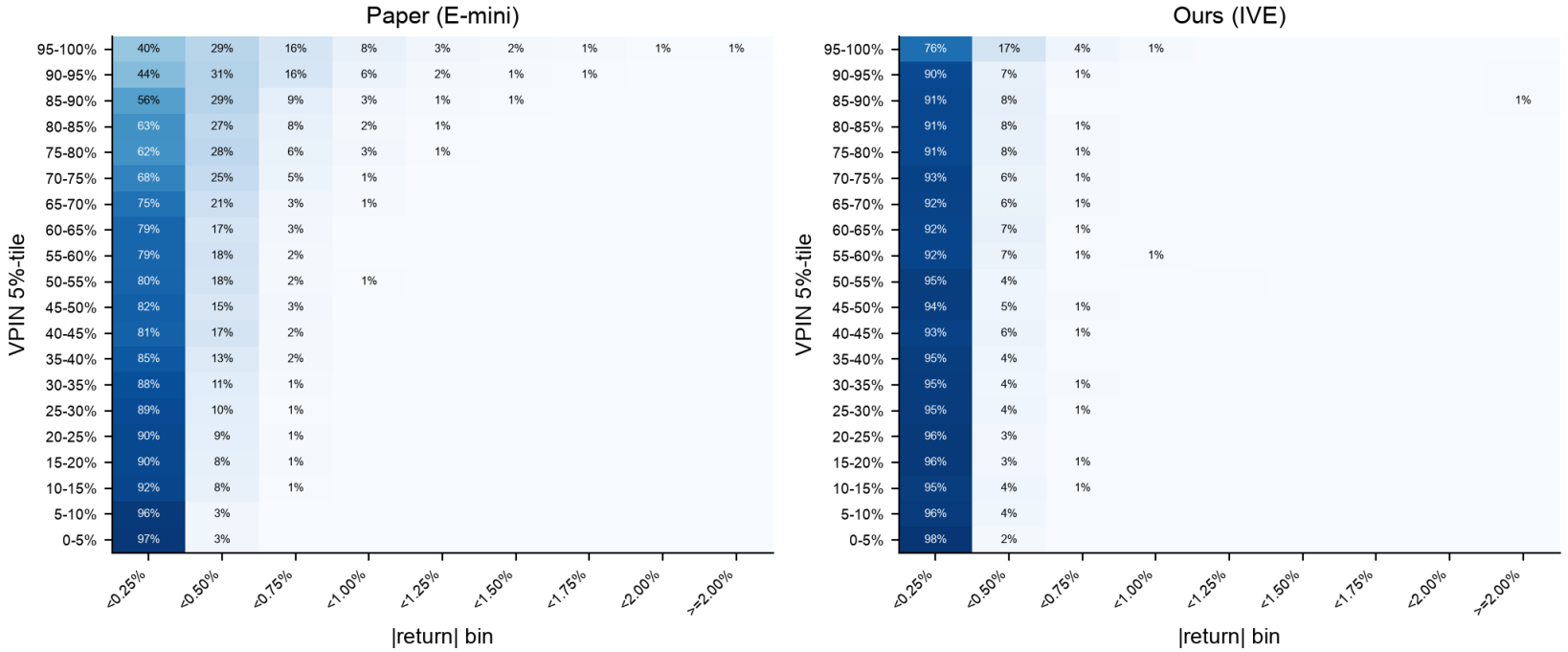
ours **0.08**

Best cell on our grid: 0.10

Same sign, a fifth of the size.

Result 4: high VPIN and large moves (Tables 4 and 5)

Table 4a: P(next |return| bin | VPIN 5%-tile), (50, 250)



- Bottom-quartile VPIN → next |return| < 0.25%: paper > 90%, ours **96%**
- P(VPIN in top quartile | next |return| > 0.75%): paper > 84%, ours **51%**
- Large move while VPIN stays in its top quartile (Table 5): paper **79%**, ours **60%** at (10, 350)
- **Low VPIN = calm reproduces; the warning value is weaker.**

Reproduction scorecard

Claim	Paper (E-mini)	Ours (IVE)	Verdict
VPIN is stable to the starting point (Fig. 5): std / mean across 1,000 starts	6.5%	3.7%	reproduced
Volume time is closer to normal than clock time (Table 1): statistics that improve, 50 / 100 draws a day	4 of 4 / 4 of 4	3 of 4 / 4 of 4 (without 6 May 2010 and overnight changes)	reproduced
CDF(VPIN) > 0.9 at least 2 hours before the flash crash (Fig. 4), BVC 1-minute bars	≥ 2 h	4.9 h	reproduced
... with BVC 10-second bars	≥ 2 h	only during the crash	not reproduced
... with Lee-Ready trade-by-trade classification	≥ 2 h	never (max CDF 0.85)	not reproduced
Low VPIN (bottom quartile) → next |return| < 0.25% (Table 4a)	> 90%	96%	reproduced
Corr(ln VPIN, next |return|) at (50, 250) (Fig. 10)	0.40	0.08	much weaker
P(VPIN in top quartile | next |return| > 0.75%) at (50, 250) (Sec. 5.2)	> 84%	51%	weaker
P(move > 0.75% while VPIN stays in the top quartile) at (50, 250) (Fig. 14)	51.8%	33.6%	weaker
... at (10, 350) (Table 5)	78.6%	59.7%	weaker

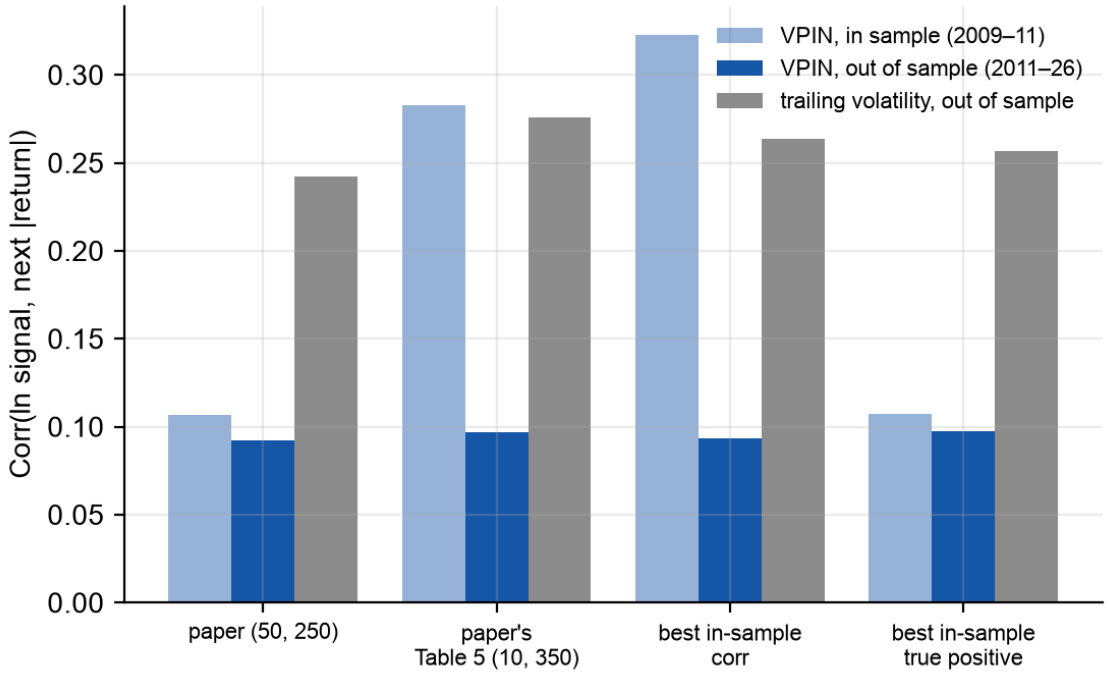
How could these results be overfit?

In the paper

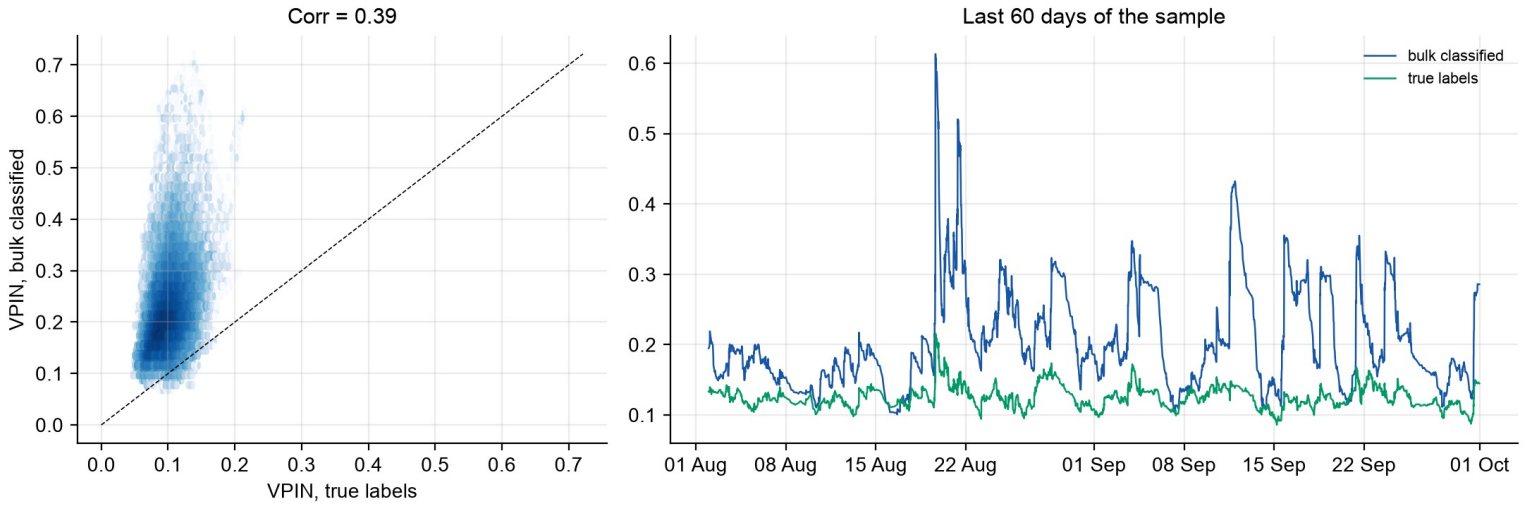
- The CDF of VPIN is built from the **whole sample**, future included.
- Bucket size and window are picked by **scanning grids** on the same data (Figs. 10, 12, 14); Table 5 reports the best cell.
- **No out-of-sample** period. One flash crash = one observation.

Our check

- Real-time VPIN: σ, V and CDF from **past data only**; parameters frozen in 2009–11, tested on 2011–26.
- Best in-sample true-positive pair (20, 500): **92%** → **45%** out of sample (bottom quartile: 47%).



Extension 1: is bulk classification right? A ground-truth test



- Bars: BVC gets the sign right for **79%–83%** of volume.
- Buckets: BVC inflates |imbalance| **2.2×**; the two VPINs correlate only **0.39**.

On 5 days with every trade

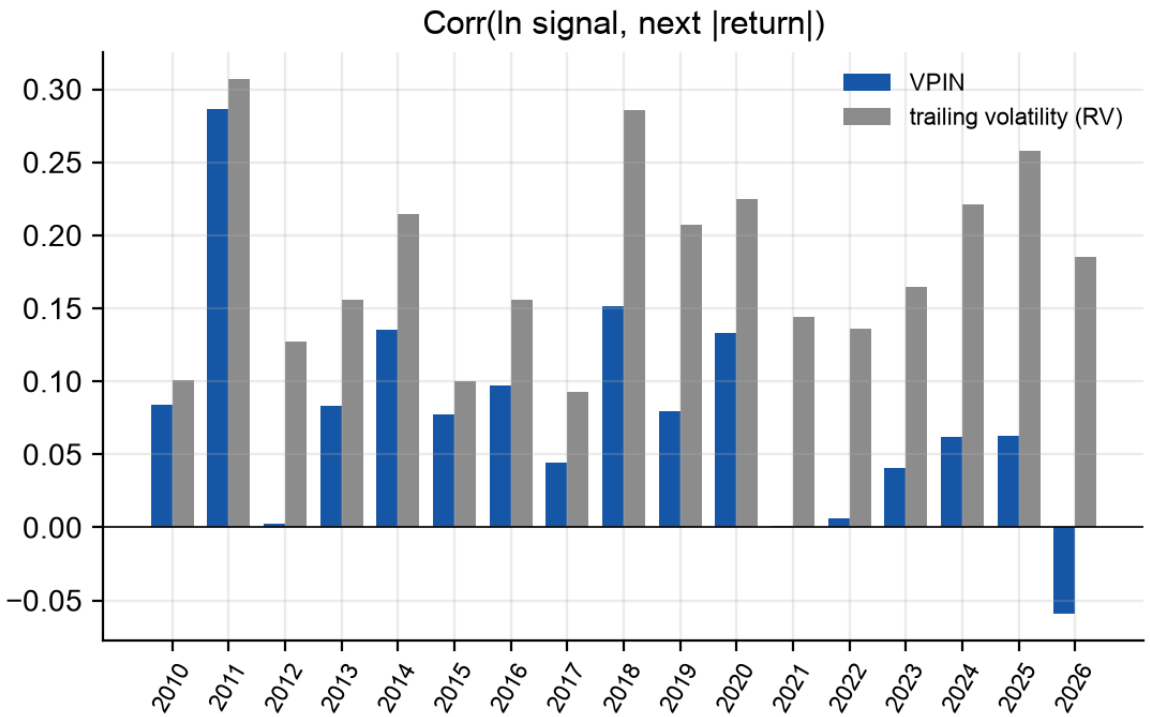
- Tick rule: **93%–97%** of volume classified right.
- Bucket imbalance vs truth: tick rule **≥ 0.99**, BVC **0.51–0.74** (1-min bars).
- Caveat: Binance aggregates the fills of one order, which helps the tick rule.
- **The paper's claim that bulk beats trade-by-trade classification does not hold here.**

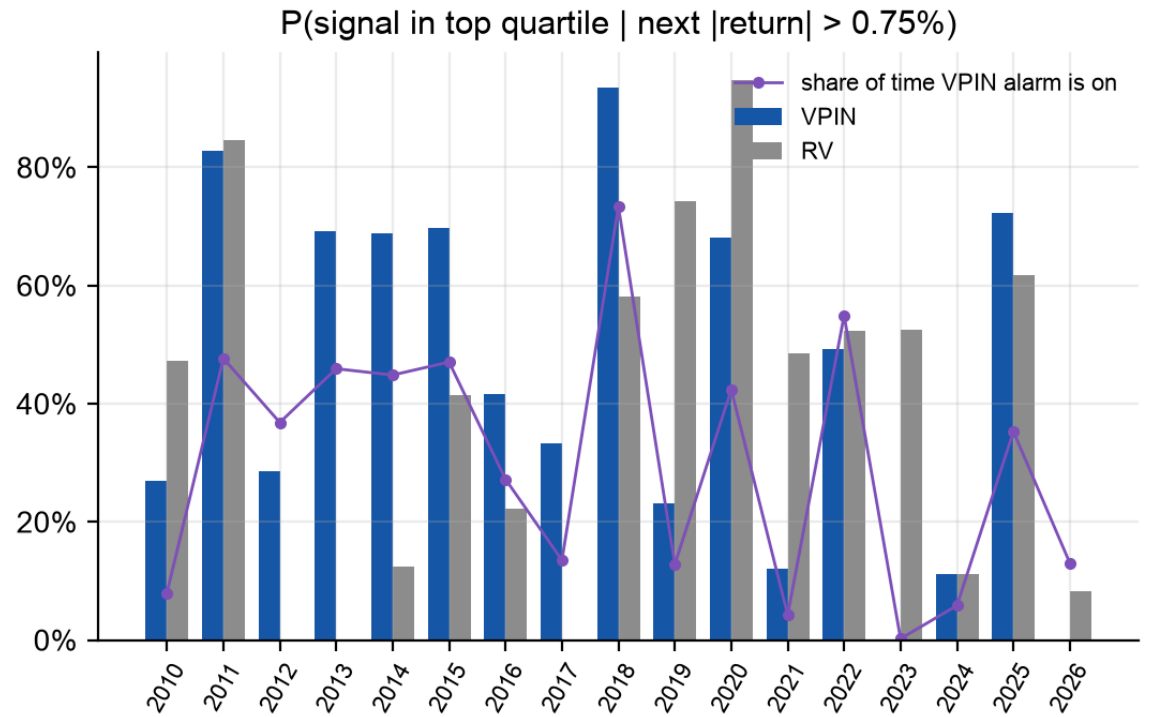
Extension 2: is VPIN more than volatility?

BTCUSDT, 2020–26 (50 buckets/day, n = 50)	Corr with next |return|	P(top quartile | move > 0.75%)	Time in top quartile	Lift
VPIN (bulk classified)	0.13	41%	27%	1.5×
VPIN (true labels)	-0.12	17%	29%	0.6×
RV (trailing volatility)	0.36	40%	12%	3.3×

- VPIN from **true labels** is negatively related to the next move (-0.12); bulk-classified VPIN is positive (0.13); **trailing volatility wins** (0.36).
- BVC imbalance moves with the bucket's own |return| (0.44 vs 0.29 for the truth): BVC builds volatility into VPIN.
- Regression with volatility and trade-intensity controls (Newey–West): VPIN adds ΔR^2 = **0.0010** on BTC, **0.0025** on IVE out of sample.

Extension 3: real-time VPIN out of sample, IVE 2011–2026





• Paper's (50, 250), out of sample: corr VPIN **0.09** vs trailing volatility **0.24**; lift **1.6×** vs **2.8×**.

Extension 4: crashes after the paper

Market	Day	Largest 1-hour fall	VPIN alarm	VPIN alarm, true labels	Volatility alarm
BTC	2021-05-19	-22.9%	on	off	off
BTC	2024-08-05	-8.9%	off	off	off
BTC	2025-10-10	-11.6%	off	on	off
IVE	2011-08-08	-3.2%	on	–	on
IVE	2015-08-24	-5.1%	on	–	on
IVE	2018-02-05	-2.7%	on	–	off
IVE	2020-03-16	-4.6%	on	–	on
IVE	2024-08-05	-1.0%	on	–	off

Hits vs how often the alarm is on

- IVE, VPIN: on at **5 of 5** falls; on **15%** of all the time.
- IVE, volatility: **3 of 5**; on 9% of the time.
- BTC: VPIN **1 of 3**, true-label VPIN **1 of 3**, volatility **0 of 3**.
- **Few events: suggestive on IVE, nothing on BTC.**

Critique and next steps

Critique

• Bulk classification **builds volatility into VPIN**; with true labels the predictive link disappears.

• Whole-sample CDF and grid-searched parameters: **look-ahead and selection** with no out-of-sample test.

• Main evidence is **one event** (6 May 2010); on IVE the warning depends on bar size and classifier.

• No baseline: the paper never compares VPIN with **plain volatility**, which wins in our tests.

• The authors hold a **patent** on VPIN and disclose a financial interest.

With three more months

• E-mini futures with **exchange aggressor flags** (e.g. CME MDP 3.0): the paper's market, with ground truth.

• VPIN as a **filter for a market-making strategy**: does it reduce adverse-selection losses after costs?

• Correct the grid search for **multiple testing** (White's reality check, deflated statistics).

• Combine with **HAR volatility forecasts** and test VPIN's incremental value out of sample.

Repository: how to run it

```
src/vpin/
  data.py          IVE ticks, BTC klines and trades; cancelled trades
  bars.py          time bars, within-day price changes
  classify.py      BVC, tick rule, Lee-Ready
  buckets.py       equal-volume buckets
  metric.py        VPIN, full-sample and real-time CDF
  analysis.py      Tables 1–5, Figures 10–14
  realtime.py      no-look-ahead pipeline
  paper_numbers.py  the paper's numbers with pages
scripts/  download, reproduce, extensions, report, slides
tests/    unit tests (incl. Table 1 formula check)
DEVIATIONS.md  every deviation and silent decision
```

Run

pip install -r requirements.txt

pytest -q

python scripts/download_kibot.py

python scripts/download_binance.py

python scripts/reproduce.py

python scripts/extensions.py

python scripts/make_report.py

Free data: 942,321 IVE trades in the paper's period; BTCUSDT 2020–2026.


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